

Multi Rater Modelling

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Outline

Motivation

Label fusion techniques

Method wrt Disagreement

Soft vs Hard on Training/Validation

Impact on Calibration

Label Fusion Choice & Takeaways

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Motivation

Data Uncertainty

In Medical Imaging, there is inherent ambiguity in the annotations → aleatoric uncertainty

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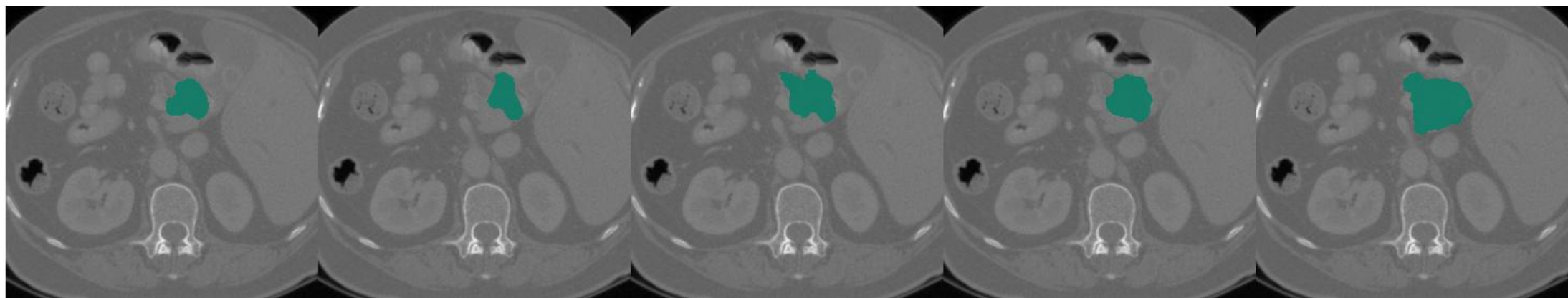
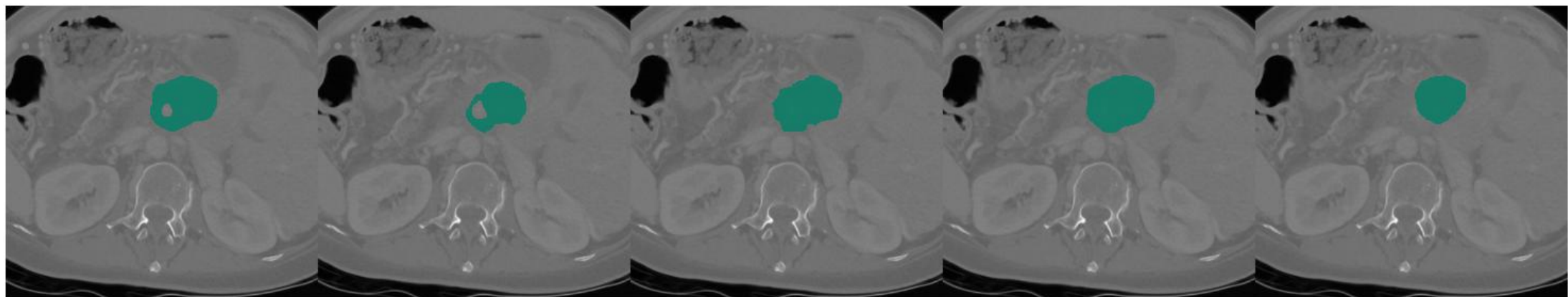
Interrater Variability

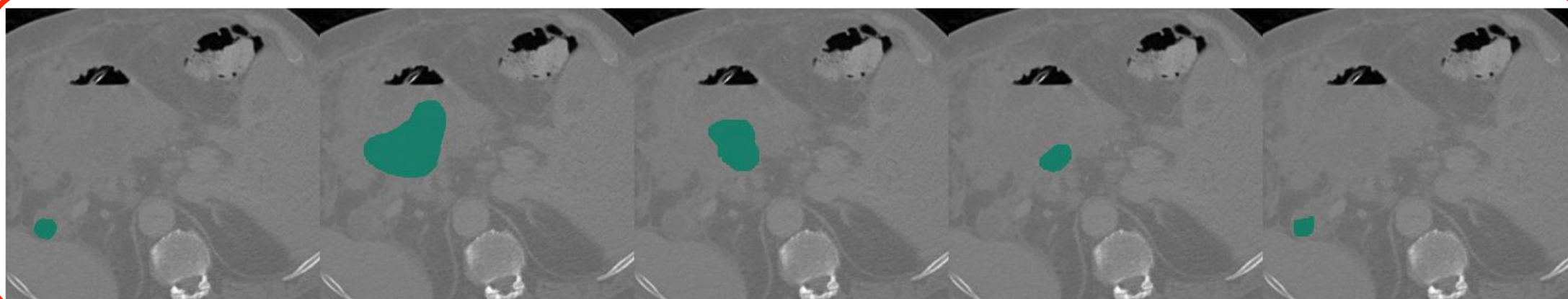
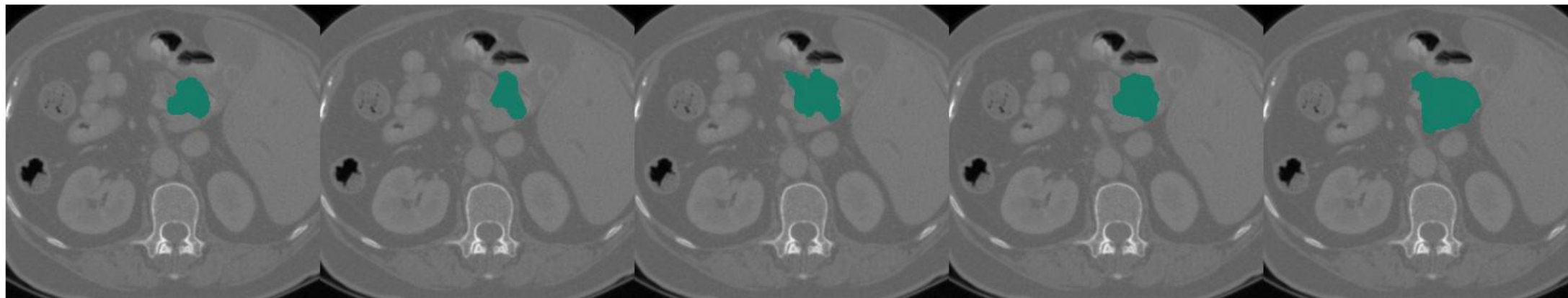
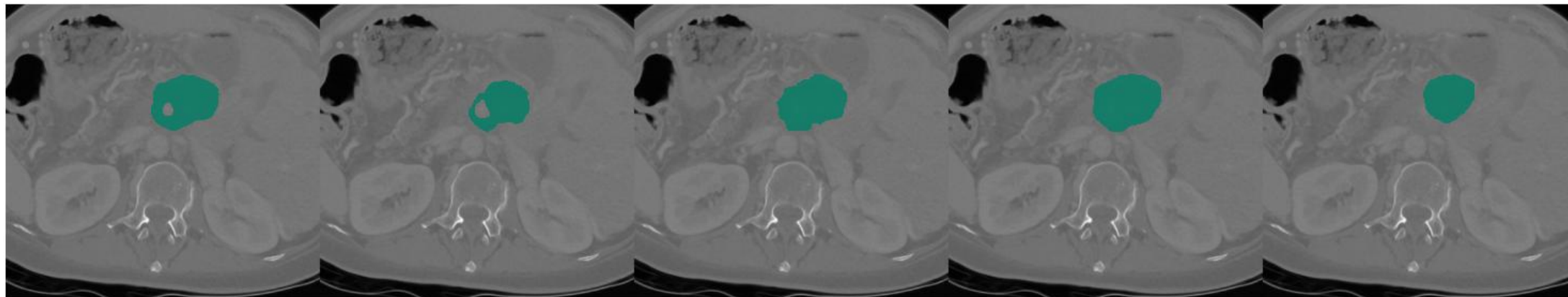
Disagreement is common – ill-defined boundaries (e.g. Tumor margins)

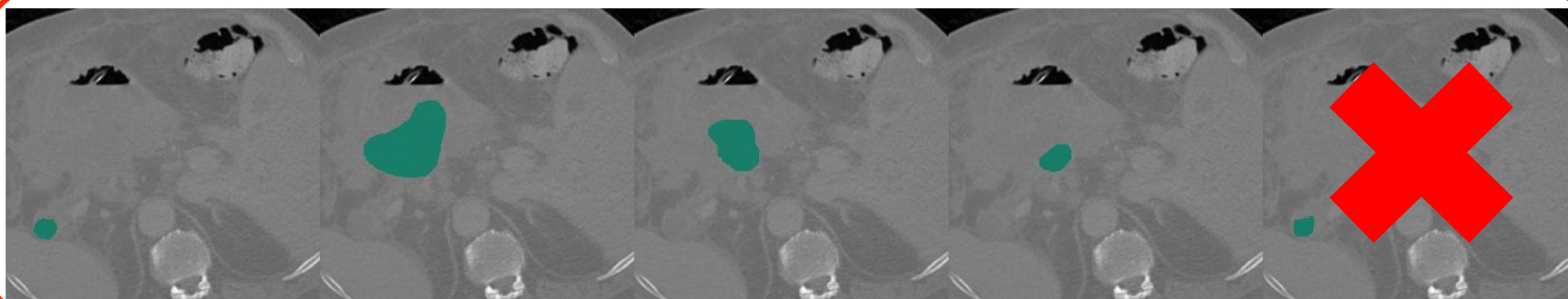
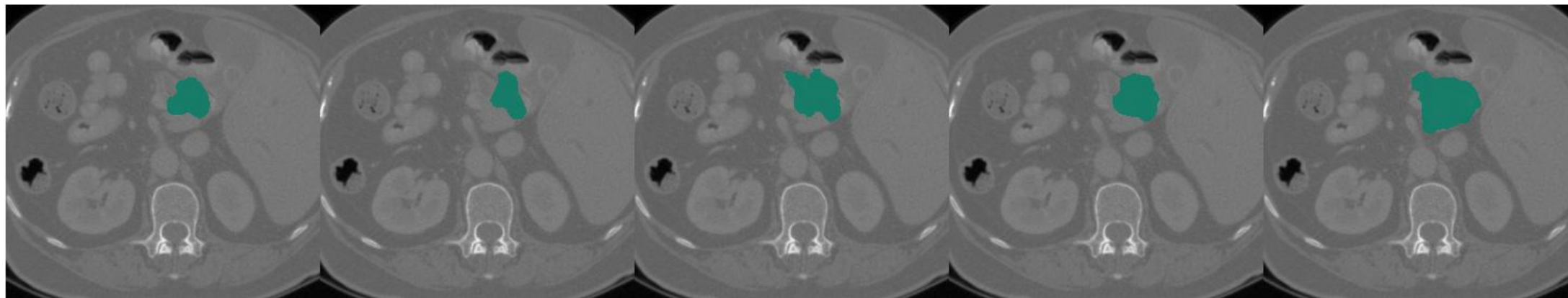
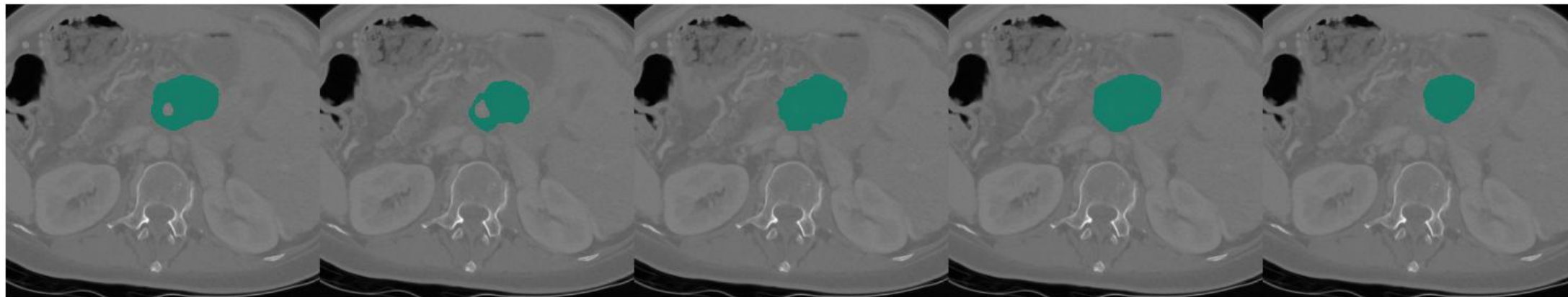
↓

Exploit this variability

Depending on the disagreement, different approaches might be required







Motivation

Naively choosing one annotation (either raw or created from others) can bias training and evaluation.

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How do we merge, model or exploit multiple labels?

What is the impact on training and validation?

How does this affect calibration and reliability?

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Label Fusion Choice & Takeaways

Label Fusion – Overview

Naive Extremes

Simple
Consensus

Random
Sampling

Soft Labels

Probabilistic
Modeling
(STAPLE)

Other Methods

Label Fusion – Naive Extremes

- Union, “captures everything”
 - High sensitivity, low specificity
 - Oversegment
- Intersection, “only agree when sure”
 - Low sensitivity, high specificity
 - Undersegment



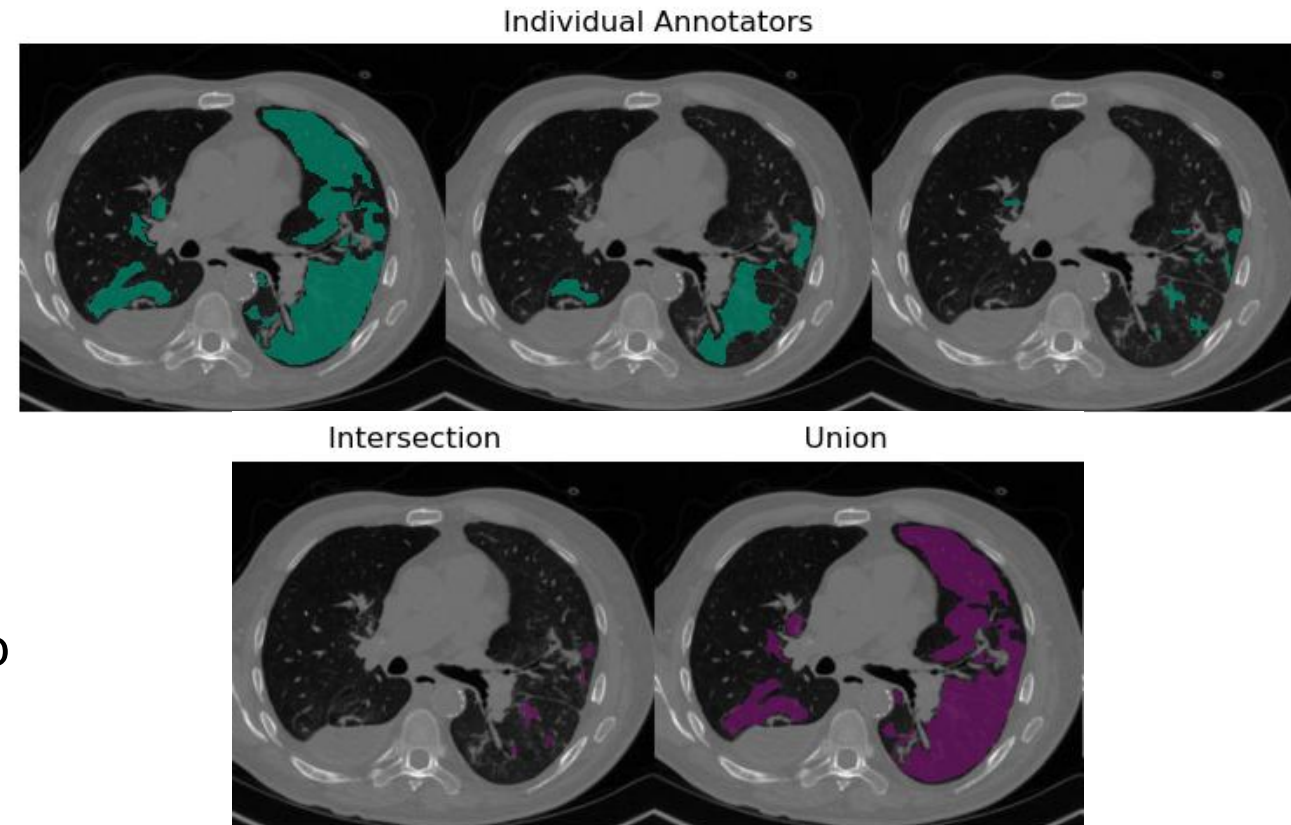
Easy to understand/implement



Oversimplify uncertainty → either too permissive or too conservative



Often used as baselines



Label Fusion – Simple Consensus

- Majority Voting (MV)
- Median = MV
 - Binary Cases
 - Odd number of raters



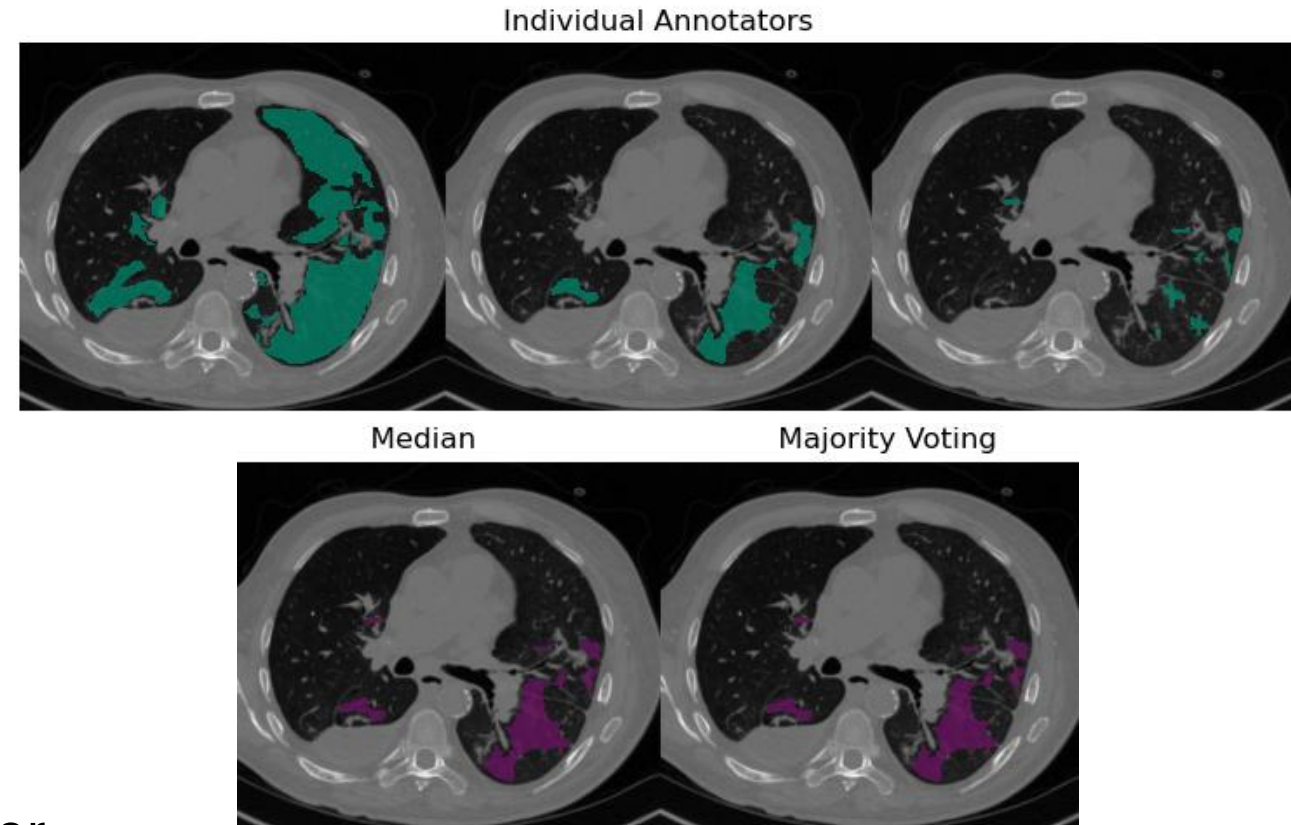
Robust to outliers



Still a hard label → ignores degree of uncertainty



Weighted voting → acknowledges rater expertise (e.g. juniors versus seniors)



Label Fusion – Random Sampling

- Instead of collapsing raters → sample one rater's label at random each epoch/iteration
- Preserves diversity across raters
- Forces model to “see” all annotation styles over training



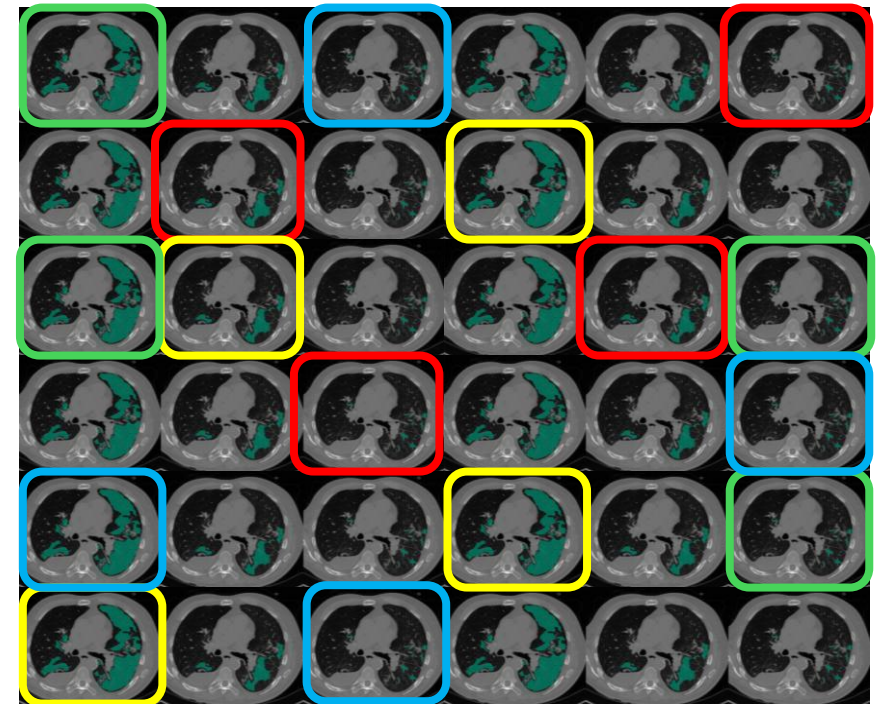
Simple, no fusion required
Effective when disagreement is high



Adds variance → requires long training to
balance exposure to all raters



Often used as a regularizer → prevents overfitting
to a single GT



Label Fusion – Soft labels

- Soft Consensus: Instead of forcing a hard “truth,” use **soft probabilities** (fraction of raters agreeing).
- Can add **spatial smoothing** → uncertainty higher near boundaries.



Uncertainty-aware labels

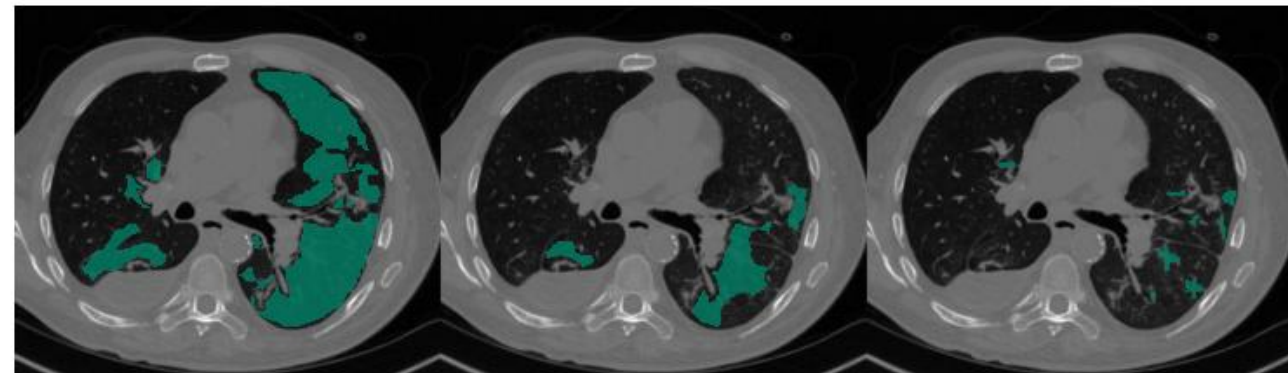


Harder evaluation → soft compatible metrics



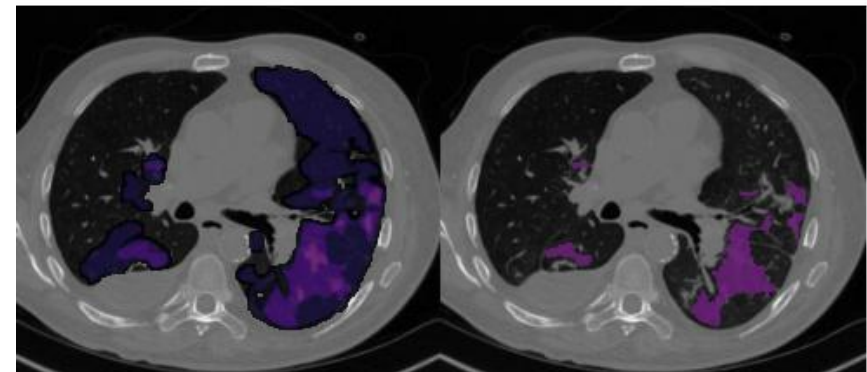
Closer to the **real distribution of expert uncertainty**, can help calibration

Individual Annotators



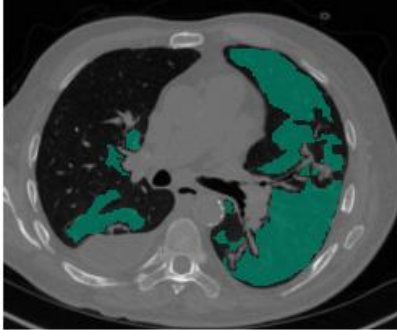
Soft-Consensus

SVLS

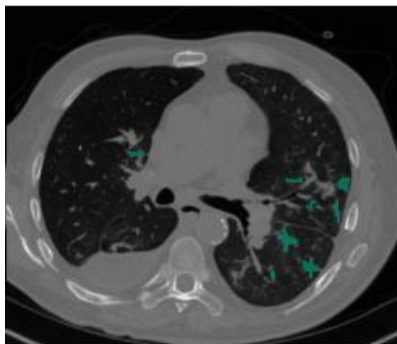
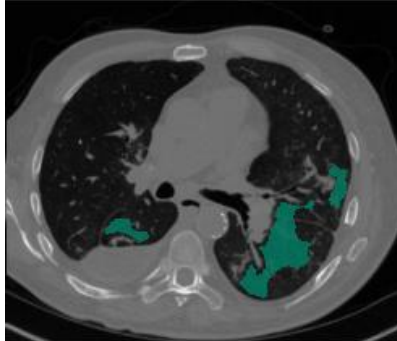
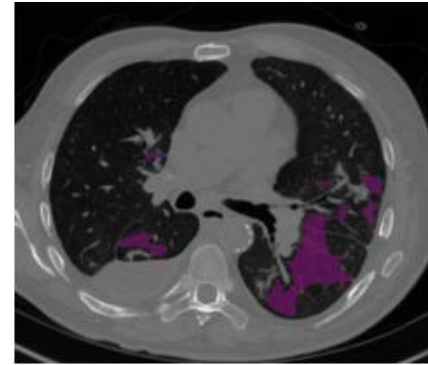


Spatially Varying
Label Smoothing

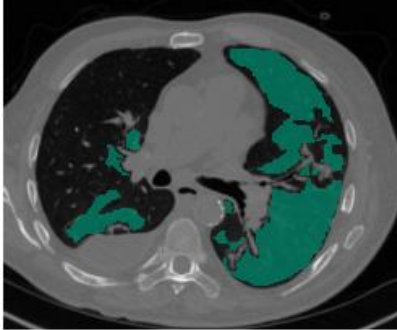
Label Fusion – STAPLE



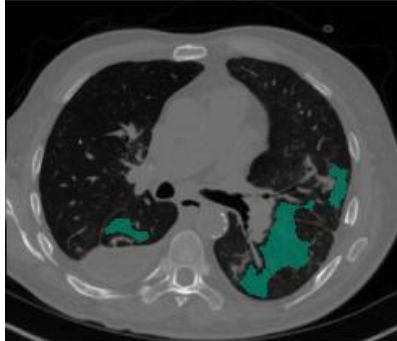
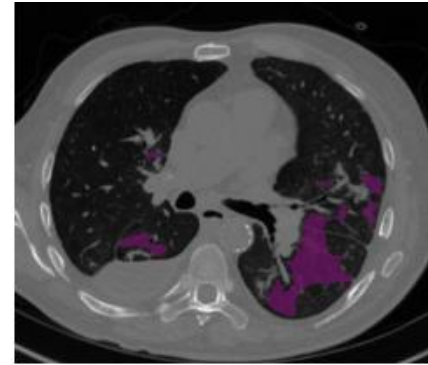
Initialize: guess
true mask
(e.g. Majority
voting)



Label Fusion – STAPLE

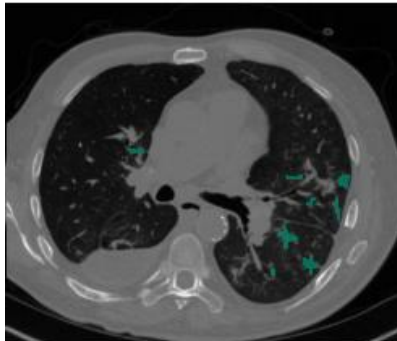


Initialize: guess
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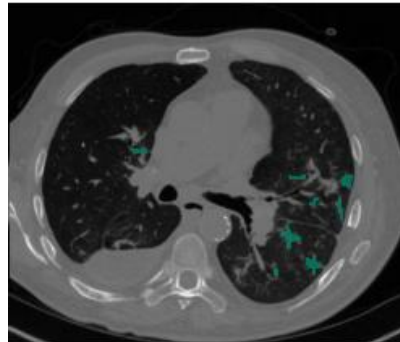
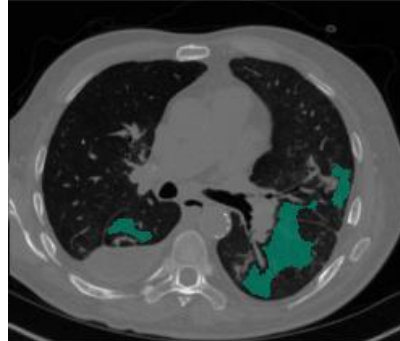
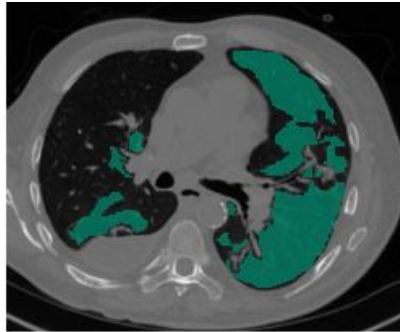


E-step: given sens/spec,
compute probability
each voxel is “true” →
probabilistic map.

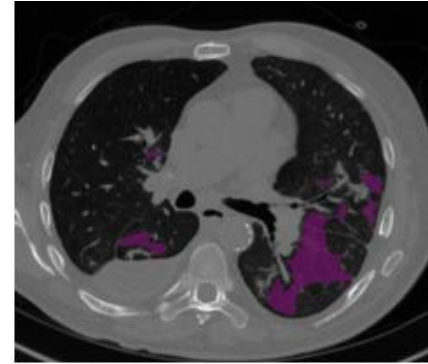
M-step: given
current “truth,”
update sens/spec for
each rater.



Label Fusion – STAPLE



Initialize: guess true mask (e.g. Majority voting)



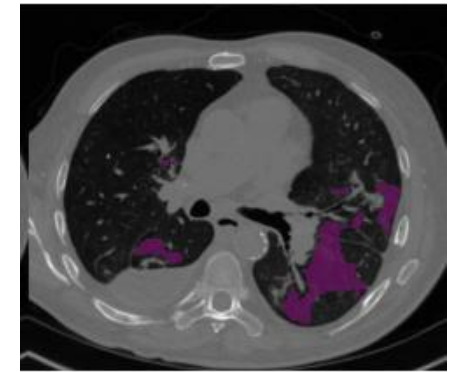
E-step: given sens/spec, compute probability each voxel is “true” → probabilistic map.

M-step: given current “truth,” update sens/spec for each rater.



When converges

STAPLE



Label Fusion – STAPLE

- Simultaneous Truth And Performance Level Estimation (STAPLE)
- Learns both a latent “true mask” and rater reliabilities



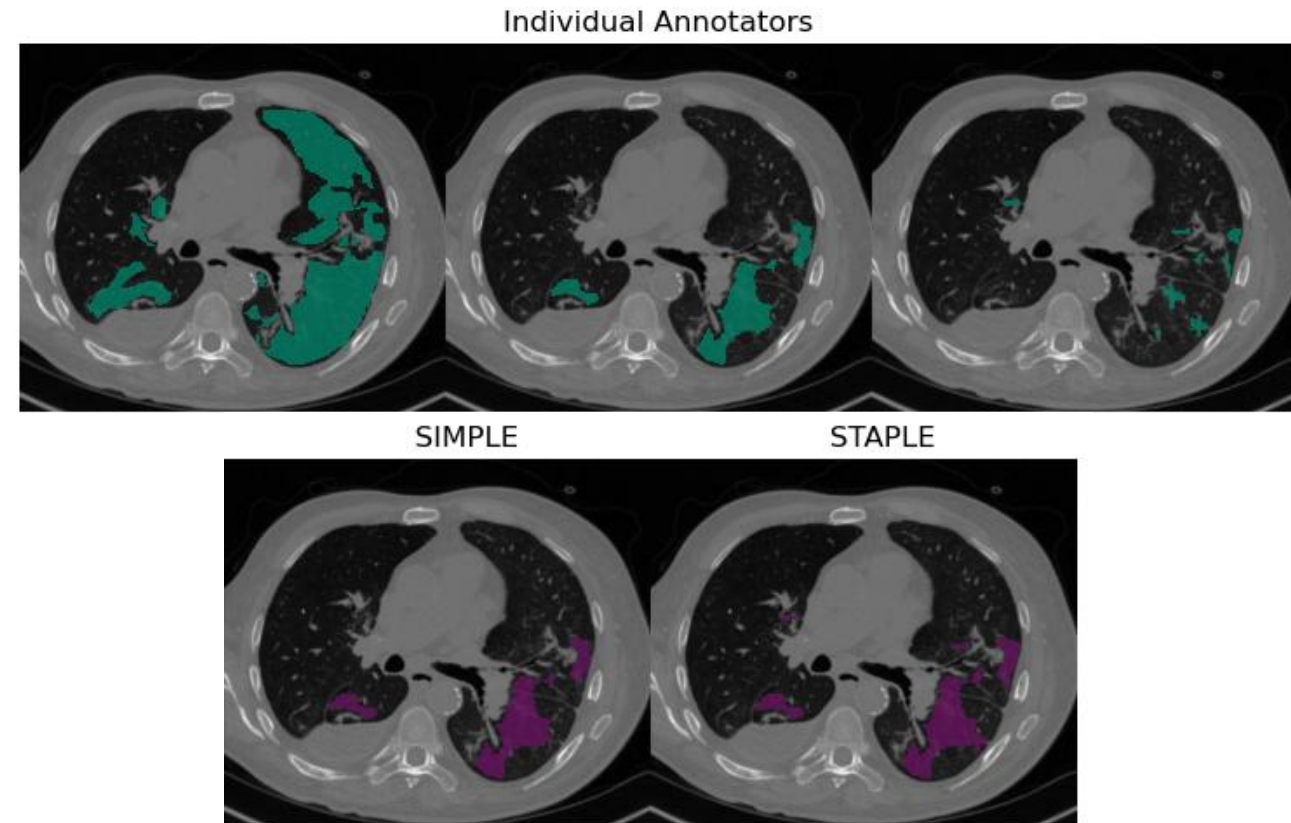
Accounts for **individual rater reliability**



Assume independence, voxel-wise independence may break for segmentation



Acknowledge raters are noisy observers



Label Fusion – Other Methods

- **DL-based label fusion:** CNNs that learn to combine rater maps into consensus
 - Small network learns to fuse multi-rater inputs into a consensus map
- **Multi-Annotator Training** – e.g. loss functions that compare predictions to *all raters* instead of a fused mask
- **Bayesian / Uncertainty-Aware Networks** – explicitly predict per-voxel uncertainty calibrated to rater disagreement.



Can incorporate spatial context & rater-specific patterns
Directly train model with uncertainty included



Require more data and careful design → harder to interpret and validate than classical methods



These are more flexible but less standardized → ongoing research

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Label Fusion Choice & Takeaways

Method wrt Disagreement

Method	Low disagreement (raters mostly agree)	High disagreement (raters split)
Union	$\approx$ Majority/Intersection $\rightarrow$ stable	Over-segmented, very sensitive
Intersection	$\approx$ Majority $\rightarrow$ stable	Under-segmented, overly strict
Majority Voting	Stable, near ground truth	Hides minority opinions $\rightarrow$ biased
Random Sampling	$\approx$ Majority $\rightarrow$ stable	Preserves diversity, robust
STAPLE	Works well $\rightarrow$ estimates reliabilities	Still OK, but limited by independence assumption
Soft Labels	Almost same as hard mask	Best at reflecting real uncertainty

When disagreement is low $\rightarrow$ all methods converge, choice doesn't matter much.
When disagreement is high $\rightarrow$ choice of fusion method **critically changes the "truth"** the model learns.

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Soft vs Hard on Training/Validation

Hard Labels		Soft Labels
Training	 • Faster convergence • Clear GT target • Simple loss functions (cross-entropy, Dice)	• Let the model learn uncertainty and variability • Reduces overconfidence → lead to better calibration • Can improve generalization in ambiguous regions
	 • Forces model to treat all disagreements as certain • Overconfidence / poor calibration • Can overfit to arbitrary consensus → implicitly assume certainty in one GT	• Loss functions compatible with continuous labels • Slower convergence if uncertainty high • Choice of smoothing parameters matters
Validation	 • Easy evaluation metrics (Dice, IoU) • Easier interpretability by clinicians	• Can compare model probabilities directly to rater agreement • Better reflection of inherent ambiguity
	 • Metrics hide uncertainty (single number) • Apparent performance depends strongly on fusion method	• Harder to benchmark with standard metrics • Need thresholding or probabilistic metrics • May show lower peak Dice even if better calibrated
Best Use Case		
Low disagreement or when binary GT is required for deployment.		High disagreement or when uncertainty estimation / calibration is critical.

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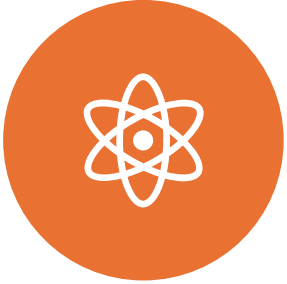
Method wrt Disagreement

Impact on model Training and Validation

Impact on Calibration

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Fusion Choice



Training Target

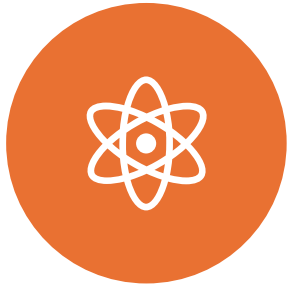


Model Confidence



Calibration Quality

Impact on Calibration



Fusion Choice

Hard Labels:

- Treat all voxels as certain
- Model tends to become **overconfident in ambiguous areas**
- Calibration curves show underestimation of uncertainty

Soft Labels:

- Encode rater disagreement in the target
- Encourage model to **predict probabilities matching rater agreement** → real uncertainty
- Lead to improved Expected Calibration Error

Label fusion isn't just preprocessing — it defines the **(probability) target** your model learns, which directly **shapes calibration**.

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Label Fusion Choice

	Why It Matters	Examples
Level of Disagreement	High disagreement → need richer models (STAPLE, soft labels) Low disagreement → simple consensus often sufficient.	Tumor margins vs. well-defined organs
Computational Resources & Time	Some methods (STAPLE, Bayesian models) require EM iterations or MCMC, can be slow for 3D volumes. Simple majority/union is fast .	Large CT volumes with many raters
Goal of the Model	If the model's main output is binary masks for deployment, simple fusion may suffice. If you need calibrated probabilities/uncertainty , prefer soft/probabilistic methods.	Clinical decision support vs. research prototype or other research fields
Complexity of the Problem	More complex anatomical regions or subtle findings → benefit from probabilistic/soft approaches . Well-defined , high-contrast structures → simpler methods .	Pancreatic tumor vs. lung parenchyma segmentation
Data Size & Number of Raters	Fewer raters → probabilistic methods may be less stable . More raters → benefits grow .	Small expert panel vs. crowdsourced annotations

Label Fusion Choice & Takeaways

- **Interrater variability is a valuable signal**, not just noise — it reflects real uncertainty in medical imaging. This uncertainty should be reflected in the predictions as well.
- Different **label fusion** methods offer **trade-offs** in **simplicity**, **computational cost**, and ability to **model uncertainty**.
- **Fusion choices** directly **shape** model training **targets** and strongly influence calibration, confidence, and clinical reliability.
- Choosing the right fusion strategy depends on **disagreement level**, **computational resources**, and the **intended goal** of the model — and this variability should be addressed not only during training but **also in evaluation**.
 - See spotlight presentation.

Thank you

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